XUEWU CAI

FINANCIAL ENGINEERING & MORTGAGE ANALYTICS LEADER

Clarksville, MD | 410-446-6815 | gourmetcai@gmail.com

linkedin.com/in/xuewu-cai-a8069664 | U.S. citizen

PROFESSIONAL SUMMARY

Financial engineering director and technology leader with more than 20 years across mortgage finance, quantitative research, risk management and trading systems. Experience includes enterprise forecasting and scenario analysis, market-data infrastructure and analytical reporting, documented use of generative AI tools in engineering workflows. Combines hands-on Java, Python, MATLAB and C/C++ expertise with large-scale analytics on AWS and IBM Symphony and documented financial model implementation.

CORE EXPERTISE

Finance & risk: Credit risk transfer (CRT), structured finance, mortgage analytics, portfolio risk, capital planning, ROE, CECL, DFAST, derivatives pricing, scenario analysis, and stress testing. Mortgage products: MBS, TBA, agency debt, and CMO; hedge accounting.

Leadership & delivery: People management, technical leadership, multi-project delivery, financial model implementation, and investment decision support.

Technology: Java, Python, MATLAB, C/C++, AWS, IBM Symphony, Docker, IntexCalc, DealMaker, QuantLib, FIX, Oracle, SQL Server, Snowflake, MongoDB, Spring Boot, PyTorch, JSON, VS Code, Amazon Q, Copilot, and harness engineering.

PROFESSIONAL EXPERIENCE

Freddie Mac | Director, Financial Engineering/Inv&Capital Market

May 2014 – Present | McLean, VA

- Direct development of modeling tools that combine prepayment and default models, large-scale computing on AWS and IBM Symphony, and Intex valuation analytics to assess bond performance and transaction economics, including return on committed capital.
- Lead financial engineering projects supporting STACR bond issuance and ACIS reinsurance within Freddie Mac’s CRT (Credit Risk Transfer) program, which protects more than $1.6 trillion in mortgage loans against credit losses.
- Own an enterprise forecasting platform for income statements, balance sheets, and CECL estimates, supporting annual business planning, risk appetite assessment, capital planning, and DFAST stress testing.
- Delivered deal memorandum, capital, economics, and ROE analytics for new RPL and synthetic structures, using Copilot and harness engineering to support delivery.
- Apply hands-on expertise in STACR, senior/subordinate structures, ACIS reinsurance, and IntexCalc, DealMaker, and CDI structures to support valuation and model implementation.
- Oversee portfolio execution and risk analytics covering duration, convexity, option-adjusted spread (OAS), key rate durations (KRDs), capital benefits, post-transaction capital, and return on equity across the CRT portfolio.
- Manage the transition from LIBOR to SOFR within CRT modeling tools and analytics.
- Use generative AI tools, including Copilot, extensively to streamline engineering workflows and improve delivery efficiency.

PROFESSIONAL EXPERIENCE — CONTINUED

Campbell & Company | Research Engineer and Engineering Roles

March 2003 – April 2014 | Washington, DC–Baltimore Area

Research Engineer, Portfolio and Risk Team | 2009 – 2014

- Developed volatility and correlation forecasting models in MATLAB using daily, intraday, implied volatility, and tick data.
- Applied statistical methods, Monte Carlo simulation, and bootstrapping to investigate indicators of market stress, portfolio drawdowns, regime changes, and correlation breakdowns using MATLAB, C/C++, and Java.
- Designed and implemented internal risk reporting covering factor analysis, scenarios, stress tests, duration, and margin to support Investment Committee decisions.
- Produced regulatory and industry reporting for SEC, CFTC, and OPERA requirements, as well as marketing and compliance analyses, using MATLAB, Java/StyleReport, and Panopticon.

Quantitative Finance Engineer | 2006 – 2008

- Led development of a Java-based intraday cash equity trading system using the Caché database; built time-series data infrastructure and market data feeds in MFC/C++.
- Designed a Java connectivity layer that unified MATLAB access to production databases, including Oracle, SQL Server, Caché, and QADirect.
- Built an FX options pricing, simulation, and risk engine using Visual C++, Boost, and QuantLib.

Senior Software Engineer | 2003 – 2006

- Led architecture and development of a multithreaded, fault-tolerant electronic trading platform using Java, Portware, and FIX, supporting concurrent strategies, orders, and multiple asset classes.
- Supported more than $3 billion in capital on the trading platform and helped save tens of millions of dollars in slippage costs.

Aquilent | Senior Software Engineer

June 1999 – February 2003

- Built Java/Linux sensor integration systems, including embedded Linux deployments, JNI hardware interfaces, serial communications, XML instrument definitions, and dynamic sensor discovery.
- Designed server-side satellite data management and visualization components, XML data models, and a JavaScript-based model equation engine.
- Developed Apache/Tomcat visualization portals, session administration tools, and an RMI socket solution enabling callbacks through enterprise firewalls.

EDUCATION & PROFESSIONAL DEVELOPMENT

University of Maryland Baltimore County — Master’s Degree, Computer Science

Shanghai Jiao Tong University — M.S., Computer Science & Engineering

Dalian University of Technology — Bachelor’s Degree, Applied Mathematics

Professional development: CAIA (Chartered Alternative Investment Analyst) Level I

LANGUAGES

Chinese: native or bilingual proficiency | English: full professional proficiency